

Robust optimal self-scheduling of potable water and power producers under uncertain electricity prices

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HIGHLIGHTS

- Robust optimization based self-scheduling of fresh water and power producers is presented.
- No need to probability density and membership functions for modeling price uncertainty.
- Both risk-neutral and risk-aversion generation schedules are optimally obtained.
- Sensitivity of water-energy nexus schedules to budget of uncertainty is analyzed.

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ABSTRACT

Recently, seawater desalination and power generation units are optimally co-scheduled because of inter-connection between electricity and water networks. A typical water-energy hub grid consists of the conventional thermal power plants and the combined water and power (CWP) generating units. In the CWP plants, the waste heat of the flue gases exhausted from the power generation process is utilized for water treatment. If the water and power generation company aims to maximize its daily profit, the electricity price uncertainty will affect the optimum operating point of the generating units. In other words, the fluctuations of the energy prices cause the power generation patterns of thermal and CWP units to change. Hence, this paper implements a robust optimization strategy on water-power nexus model to handle the uncertainty of the electricity price with no need for its probability distribution and membership functions. The lower and upper bounds and the forecasted prices are used for solving the robust mixed-integer non-linear program and making the risk-averse decisions against the uncertainty. It is indicated that the proposed approach is suitable for the price taker water-power cogeneration companies, which seek the optimal schedule of their thermal and CWP units for a certain operating period.

1. Introduction

In recent years, 40% of the global population face with the water scarcity, environmental pollution, and the limited capacities of the fossil fuels based power generation stations [1]. The higher energy consumption in developing countries such as Iran has caused the huge volume of the pure water is utilized by the conventional steam turbine power plants [2,3]. In other words, the desalted water is utilized in the energy procurement processes such as thermally driven power plants and the electricity is consumed by the gross water treatment units and the fresh water distribution equipment such as pumps [4,5]. It is forecasted that the global water and electricity demands are supposed to

increase by 50% and 80% in 2050 [6,7]. In the tropical areas, the water scarcity and the energy crisis usually increase due to the temperature rise, which may lead to the load-generation imbalance and the forced outages of power and potable water distribution lines [8]. Therefore, the water and energy networks should be co-operated in order to minimize the fuel cost of power only units and cogeneration plants [9,10].

Many researchers have proposed the innovative designs of the renewable energy resources based interconnected water and power generation systems and optimized their performance from the energy, exergy, economic, environmental, efficiency points of view. Jabari et al. [9] presented a water-heat-power nexus model and optimized its

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Nomenclature

Subscripts and superscripts

i	thermal power plant
j	combined water and power generation unit
t	operation time[h]

Symbols

a_i, b_i, c_i	constant factors related to the fuel cost of thermal unit i
$a_j, b_j, c_j, d_j, e_j, f_j$	fuel cost coefficients of CWP unit j
C_{cwp}	the operation cost of the gas-fired CWP plants during the sample day (\$)
$C_{i,ini}^k/C_{i,fin}^k$	the initial/final fuel cost of the thermal unit i for generating $P_{i,ini}^k/P_{i,fin}^k$ (\$)
FC_p, STC_p, SDC_p	the fuel cost, start-up cost, and the shut-down cost of the thermal power only units over the study horizon (\$)
P_g^t	the total electrical power generated by the thermal power plants and the CWP units (MW)
$P_{j,t}$	the output power of the co-producer j at hour t (MW)
$P_{i,ini}^k/P_{i,fin}^k$	the initial/final value of the power generation of unit i in block k (MW)
$P_{i,t}$	the output power of the thermal power plant i at hour t

P	(MW)
$-i,t/\bar{P}_{i,t}$	the lower/upper bound of the power generation region for the thermal unit i at hour t (MW)
P_j^{min}/P_j^{max}	the minimum/maximum values of the power produced by the CWP unit j (MW)
R	the total money obtained from selling energy to the local electricity network (\$)
st_i and sd_i	the start-up cost and the shut-down cost of the power only unit i at operating time interval t (\$/h)
SU_i/SD_i	the start-up/shut-down rate of unit i at hour t (MW/h)
$W_{j,t}$	the output fresh water of the co-producer j at hour t (m ³ /s)
W_j^{min}/W_j^{max}	the minimum/maximum values of the water produced by the CWP unit j (m ³ /s)
$y_{i,t}$ and $z_{i,t}$	the auxiliary binary decision variables for modeling the start-up and shut-down of unit i at hour t
Γ	budget of uncertainty
ΔP_i^k	the length of the linear section k (MW)
δ, ξ_t	dual variables of the risk-averse optimization problem
$\hat{\lambda}_t$	the maximum deviation of the actual energy price from the forecasted one at hour t [\$/MWh]
λ_t	the uncertain electricity price at time interval t [\$/MWh]
$\bar{\lambda}_t$	the forecasted value of the energy price at hour t [\$/MWh]

performance over a 24-h study horizon. In this research, it is supposed that the peak electrical demand is more than the maximum power generation bound of thermal units, combined heat and power (CHP) systems, and combined fresh water and power (CWP) generation cycles. Hence, it is infeasible to satisfy it without co-operation of thermal, CWP, and CHP units. Moreover, the pure water consumption in the benchmark hub system is higher than the maximum capacity of the desalination only units. Therefore, desalination plants and CWP units are optimally co-dispatched to satisfy this volume of the potable water. The similar concept is considered for the heat demand, which can be supplied by the CHPs and boilers. Moreover, a time-value based demand-side management strategy is implemented on heat/water/power loads to shift a percentage of each demand from peak periods to off-peak hours, flat their daily variation patterns, and reduce the fuel cost of the trigeneration grid. The economic benefit of the aggregated plug-in pure electric vehicles in transportation and energy procurement sectors is proved by solving a mixed integer nonlinear programming problem (MINLP). The uncertainty of water/heat/power demands, electricity prices, and availability of the electric drive vehicles in a 1-year study period should be incorporated to the risk-constrained long-term planning of the trigeneration systems using the time series Box-Jenkins model such as autoregressive integrated moving average (ARIMA) and differential autoregressive moving average (ARMA) tools [11]. In [12], the waste-water purification and water consumption during the energy production from the wind source are evaluated. The network-environment analysis of China's standard network demonstrated that the wind power generation system consumes lower fresh water than others. In addition, 90% of the power consumption in water systems is related to the water extraction process. Although the waste water recycling utilizes a large portion of the wind power, the compartment of the ground/surface water is beneficiary from the waste treatment. Some factors such as the old types and high corrosion of the constructed pipelines, maintenance of pumps and valves, and the mechanical damage cause the underground water system leakage [13]. To solve this issue, the pressure of the pumping stage should be arisen, which makes the water leakage worse, as well as more electricity will be consumed by the water pumps [14]. The water leakage detection tools reduce the energy requirement of the water distribution systems [15]. Authors of [16–18] proved that the ocean thermal energy

conversion based binary cycle generates the electrical power and desalted water. Authors of [19] studied the potential of the wave energy powered desalination plant to solve the potable water scarcity of the mid-latitude areas such as Canary Islands. Offshore compressed air energy storage (OCAES) [20] and undersea pumped storage system [21] could be integrated with marine, tidal and ocean wave based energy production facilities to increase efficiency and output power.

In [22], the biogas-fueled Brayton cycle is coupled with a humidification-dehumidification (HDH) desalination unit to supply the energy and fresh water demands of the industrial consumer. The uncertainty of loads and climatic condition could be modeled using the information gap decision theory to make the robust or risk-aversion and the opportunistic or risk-taker decisions and minimize the capital investment, maintenance and the operating costs of the multi-generation system [23]. In addition, bioethanol [24] and biodiesel [25] could be used instead of biogas to improve the overall efficiency of the cogeneration system. The optimal design of the grid-tied and off-grid solar PV/diesel generator/battery/inverter power system for coupling with seawater desalination unit is provided by Ghenaie et al. [26]. It is recommended to make its performance more robust against the uncertainty of the solar irradiance. For instance, molten salt thermal storage should be installed for saving heat at hot hours and discharging it for desalting water and generating power at night or cloudy sky condition [27]. In [28,29], organic Rankine cycle (ORC) is used for generating heat and power for membrane distillation and reverse osmosis cycles, respectively. The output brine of the reverse osmosis process is reheated by the surplus heat exhausted from the ORC to treat more fresh water from the membrane distillation procedure. It is found that integration of membrane distillation with ORC coupled reverse osmosis unit improves both economic saving and environmental aspects associated with natural gas utilization. In comparison with membrane distillation approach, the implementation of HDH [30], forward osmosis [31], and multi-stage flash desalination [32] to the ORC-reverse osmosis combined cycle could result in better emission footprint, efficiency, and power consumption, respectively.

Authors of [33] proved that the integration of Brayton, multi-effect desalination and vapor compression cycles using a dual-pressure heat recovery steam generator leads to produce 85 kg/s pure water, 30 MW electricity, 2 MW cool, and 1 MW heat under 39% energy efficiency for

a large-scale industrial customer. If an air source heat pump is installed for precooling the air, the Brayton cycle will consume less electrical power in the compression stage and more electricity will be produced by the gas turbine [34,35]. In [36], the brackish water reverse osmosis desalination unit is controlled by the power field oriented control strategy to protect the DC bus voltage stability against the undesired values of power and pure water utilization of the hydro-mechanical customers. Moreover, the electrical power is shared between the pumps and motors for managing the energy demand. The gas-steam combined cycle, thermochemical copper-chlorine, proton exchange membrane fuel cell, and the reverse osmosis are integrated in [37] to generate 1.7 MW power, 51.8 kg/h hydrogen, and 2.3 kg/s desalted water at 58% energy efficiency for 200 residential buildings. In this research, the flue gases exhausted from the steel melting furnace with the temperature 810 °C is used for driving system. In summer, the interconnected power system is usually overloaded. Hence, some of the large industries such as steel making companies may voluntarily or mandatory be participated in peak-clipping programs by shifting their operation period to off-peak time intervals [38,39]. Hence, the reliability of the poly-generation system will decrease due to the zero production of the flue gases. It is recommended that another auxiliary fuel is also utilized to improve its coefficient of performance when the furnace is off. Noorollahi et al. [40,41] demonstrated that the abandoned gas and oil wells as the clean and sustainable source of the heat reservoirs can drive the multi-effect desalination system as well as produce electricity and heat. They studied Ahwaz oil field in Iran and found that at least 565 m³ potable water, 1.8 MW heat and 364 kW power are produced during the sample summer day, which can mitigate the greenhouse gases, water shortage, and energy crisis in this hot and humid region. It is recommended that the molten salt thermal energy storage and the solar photovoltaic panels are linked with the ground source heat pump to

improve the thermal and energy efficiencies of the system, provide heat for HDH desalination cycle, and generate more electricity [42,43]. According to [44,45], the pumped hydro-electric storage and the cyber-physical systems such as internet of things (IOTs) are the cost-effective tools in energy management of waste water recycling, purifying and distributing stages. The nuclear fission reaction is a more reliable and renewable heat source to distillate the gross water [46]. If the heating energy extracted from the nuclear reactor is employed by the steam driven seawater desalination unit, its output electrical power will be reduced. In [47], it is demonstrated that the supercritical CO₂ Brayton cycle should be replaced with the ORC to mitigate the power decrease during the waste heat recovery. Moreover, He et al. [48] proved that the efficiency and values of power and water outputs of the ORC-HDH combined cycle should be improved by flashing steam and utilizing it for generating power. This heated brine can be used for heating the impure water, dehumidifying the humid warm air, and producing the fresh water. Because of the lower ramp up and down rates of the nuclear power stations in comparison with the steam and gas-fired ones [49,50] as well as safety factors of integral pressurized water reactor, they can coupled with thermally driven vapor compression and multi-effect distillation cycles and generate 475 kg/s fresh water from 800 kg/s seawater.

In the literature review, it is found that there are many valuable and interesting works on design, performance assessment, and optimization of water-energy nexus models. However, the robust optimization of the cogeneration grids under the uncertainties associated with water and power demands, energy market prices, and renewable energy resources have not been presented by scholars. Therefore, this paper proposes a risk-constrained optimization framework for CWP generation systems. The novelties of the present paper are as following items:

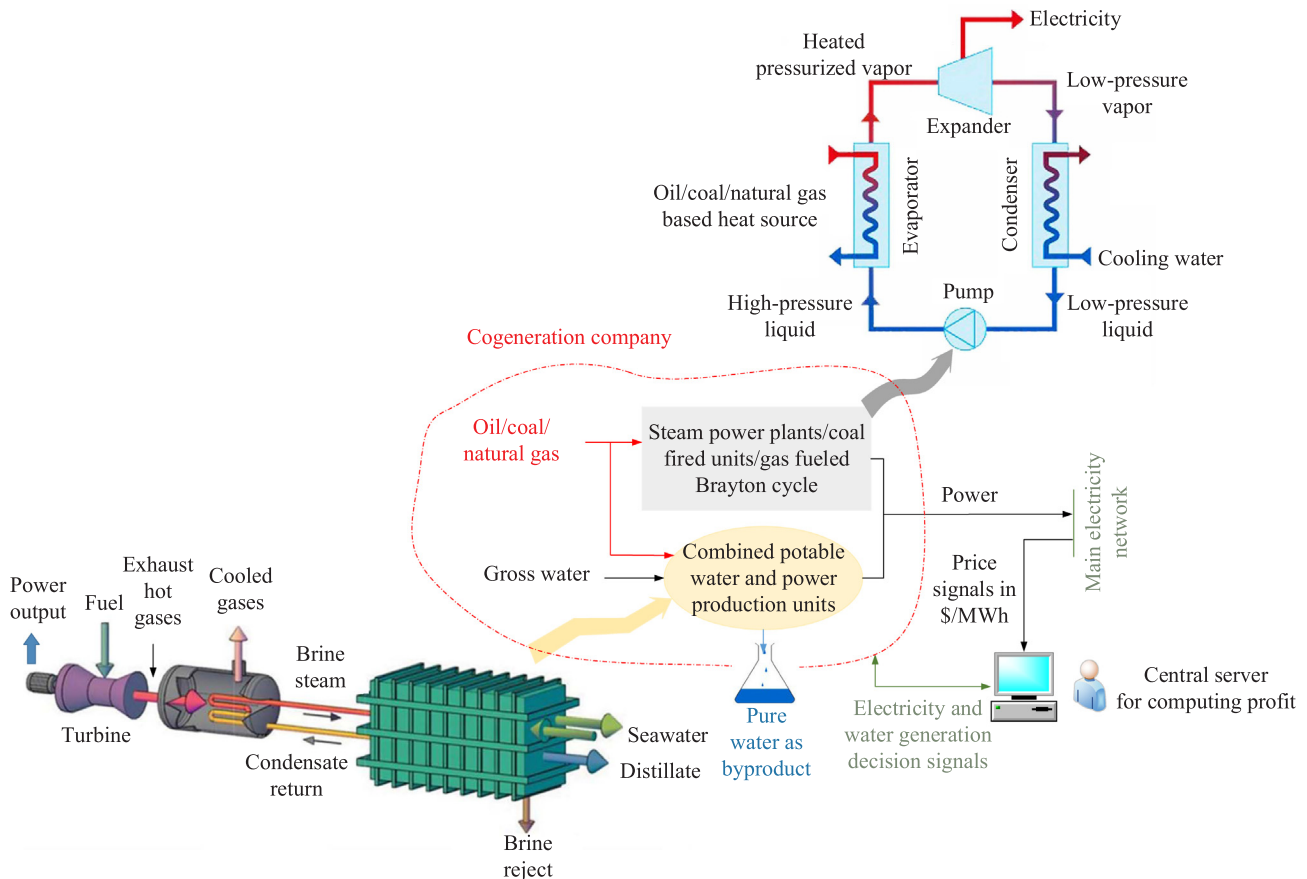


Fig. 1. The schematic presentation of the proposed energy trade between the price-taker cogeneration company and the main electricity network.

- A robust optimization technique is implemented on water-energy coupled networks. In this research, it is supposed that a typical water-power generating company (Genco) is composed of some power only plants and gas-fired thermal generating units. In the gas turbine power generation cycle, the high-temperature waste heat is exhausted from the flue gases and used for seawater purification. Therefore, these units are called CWP generation stations. The daily profit of the Genco, which is equal to the revenue obtained from selling energy minus the total operating cost of CWP units and power only plants, is maximized as the objective function. The operation cost of the thermal power plants consists of their fuel cost, start-up cost, and shut-down cost. The linearized fuel cost calculation model, ramp up and down rates of power only units, minimum up and down times of thermal power plants, generation capacity, and the power-water ratio of the CWP units are incorporated in the robust optimization problem.
- The uncertainty associated with the electricity market prices is modeled via the robust optimization algorithm. In the robust model, the lower and upper boundaries of the energy prices are used for making some stochastic scenarios. The forecasted electricity prices are firstly considered to solve a deterministic or risk-neutral optimization problem without considering uncertainty. Then, the stochastic scenarios are selected within the price interval to solve the risk-aversion robust problem.
- The sensitivity of the daily profit to the budget of uncertainty are discussed. The budget of uncertainty changes from 0 to 100% for a 1-day study horizon. For example, if the budget of uncertainty is considered to be equal to 50%, the cogeneration company maximizes its profit by considering that the actual price values in 50% of hours ($t = 1$ to $t = 12$) may be different from the forecasted ones.

The rest of the present work is organized as follows: Section 2 formulates a self-scheduling problem for CWP and thermal units. In Section 3, the robust optimization is applied on the water-energy nexus model to maximize the daily profit of the cogeneration system. In Section 3, simulations are conducted on a real case study to reveal the robustness and effectiveness of the presented approach. Then, the conclusion is provided in Section 4.

2. Problem formulation

In a combined water and energy production grid, the electricity price is affected by both generation and consumption sides. Therefore, the energy market price is a fluctuated parameter. In the literature, several scholars have used the scenario based stochastic programming approaches [51,52], Monte Carlo simulation [53], fuzzy method [54,55], artificial neural network [56] to model the uncertainty of the energy rates. The main objective of the current research is to robustly schedule the water-energy nexus model against the uncertainty of the electricity market prices using a robust optimization algorithm. According to Fig. 1, it is supposed that a water and energy cogeneration company, which composed of some thermal units and CWP plants, is linked with the main electricity grid. In the CWP production facilities, the low-grade waste heat of the flue gases exhausted from the power generation stage, is utilized for seawater desalination applications. Hence, the freshwater is considered as a byproduct. According to objective function (1), the Genco is a profit maker and desires the maximum revenue obtained from selling energy and the minimum operating cost of CWP and thermal units over a T -hour study horizon [57]. As given by (2), R refers to the total money obtained from selling energy to the local electricity network. At each time interval t , the total electrical power generated by the thermal power plants and the CWP units, P_g^t , is computed as constraint (3). The operation cost of the gas-fired CWP plants during the sample day is shown with C_{cwp} and calculated from (4). The output power and fresh water of the co-producer j at hour t , which depends on the binary decision variable $u_{j,t}$, are

represented with $P_{j,t}$ and $W_{j,t}$, respectively.

Objective function of cogeneration company

$$= \text{Max}(R - C_{cwp} - FC_p - STC_p - SDC_p) \quad (1)$$

$$R = \sum_{t \in T} \bar{\lambda}_t P_g^t \quad (2)$$

$$P_g^t = \sum_{i \in \Omega_p} P_{i,t} + \sum_{j \in \Omega_{cwp}} P_{j,t}; \forall t \in T \quad (3)$$

$$C_{cwp} = \sum_{j \in \Omega_{cwp}, t \in T} (a_j P_{j,t}^2 + b_j P_{j,t} W_{j,t} + c_j W_{j,t}^2 + d_j P_{j,t} + e_j W_{j,t} + f_j u_{j,t}) \quad (4)$$

It should be mentioned that the water and power generation capacities of the CWP producers are limited by minimum and maximum values, as inequality constraints (5) and (6). The binary decision variable $u_{j,t}$ is used to find the on and off states of the CWP units at each hour. If $u_{j,t} = 0$, the lower and upper bounds of the constraints (5) and (6) will be zero. Hence, the co-generation unit j will be off in time t . Else if $u_{j,t} = 1$, the CWP producer j will be on at hour t . Then, its water and power products will be greater than minimum and less than maximum values. where, $W_j^{\min}/W_j^{\max}$ and $P_j^{\min}/P_j^{\max}$ are the minimum/maximum values of the water and power produced by the CWP unit j . Based on (7), each co-generator can be operated at a feasible region, which is limited by its minimum and maximum power-water ratio limits [58]. Moreover, the power to water ratio should be restricted within a pre-defined interval. For example, if minimum power to water ratio is considered 4, as reported in Table 1, the power production of the CWP units is 4 times greater than their water outputs. This proves that the fresh water is selected as a byproduct. Obviously, the power and water generation capacities of the CWP units are limited, which results in a maximum power to water ratio.

$$u_{j,t} \times W_j^{\min} \leq W_{j,t} \leq u_{j,t} \times W_j^{\max}; \forall j \in \Omega_{cwp}, \forall t \in T \quad (5)$$

$$u_{j,t} \times P_j^{\min} \leq P_{j,t} \leq u_{j,t} \times P_j^{\max}; \forall j \in \Omega_{cwp}, \forall t \in T \quad (6)$$

$$R_j^{\min} \leq \frac{P_{j,t}}{W_{j,t}} \leq R_j^{\max}; \forall j \in \Omega_{cwp}, \forall t \in T \quad (7)$$

The fuel cost, start-up cost, and the shut-down cost of the thermal power only units over the study horizon are defined as FC_p , STC_p , and SDC_p , respectively. The piecewise linear model of the fuel cost function for the thermal unit i is illustrated in Fig. 2. It is assumed that the variable fuel cost function of the thermal unit i is divided into n piecewise linear sections. The initial and final values of the power generation of this unit in block k are defined as $P_{i,ini}^k$ and $P_{i,fin}^k$, respectively. where, k changes from 1 to n . For generating these amounts of electrical power, the generation company i pays $C_{i,ini}^k$ and $C_{i,fin}^k$ to purchase fuel. Therefore, the slope of the k th block, s_i^k , is obtained by dividing the

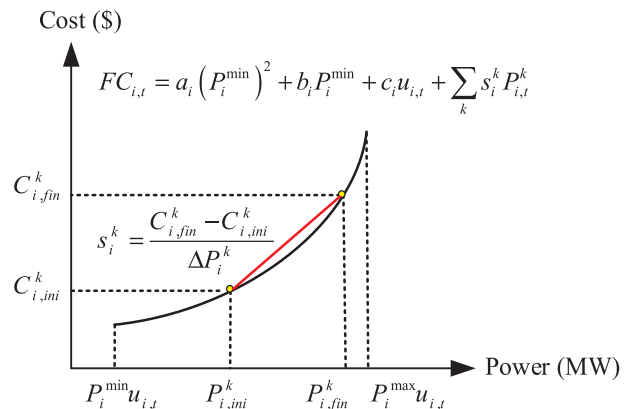


Fig. 2. The linearized model of the fuel cost function for the power only unit i .

difference of the fuel costs into the power changes from initial to final point, as computed in (8). As obvious from Fig. 2 and stated in (9), ΔP_i^k is the length of the linear section k , which represents the power changes from initial to final point of the block k . As expressed by (10), the maximum value of the power procured at time t in block k equals to ΔP_i^k and depends on its on/off status.

$$s_i^k = \frac{C_{i,fin}^k - C_{i,ini}^k}{\Delta P_i^k}; \forall k = 1: n \text{ and } \forall i \in \Omega_p \quad (8)$$

$$\Delta P_i^k = \frac{P_{i,fin}^k - P_{i,ini}^k}{n}; \forall k = 1: n \text{ and } \forall i \in \Omega_p \quad (9)$$

$$0 \leq P_{i,t}^k \leq \Delta P_i^k \times u_{i,t}; \forall k = 1: n \text{ and } \forall i \in \Omega_p \quad (10)$$

The initial and final values of the power in block k are determined using the lower generation capacity of unit i and the section length, as formulated in (11) and (12), respectively. Constraint (12) demonstrates that the power production of the thermal unit i at the end of the linear section k is equal to its product at initial point of the piecewise linear section k plus the horizontal length of the block k .

$$P_{i,ini}^k = \Delta P_i^k \times (k - 1) + P_i^{min}; \forall k = 1: n \text{ and } \forall i \in \Omega_p \quad (11)$$

$$P_{i,fin}^k = \Delta P_i^k + P_{i,ini}^k; \forall k = 1: n \text{ and } \forall i \in \Omega_p \quad (12)$$

Hence, the thermal power plant i generates the electrical power $P_{i,t}$ at hour t , which is obtained from Eq. (13). It depends on minimum power generation capacity of unit i and sum of power generated in k linear blocks.

$$P_{i,t} = P_i^{min} \times u_{i,t}^{min} + \sum_{k=1}^n P_{i,t}^k; \forall i \in \Omega_p \quad (13)$$

The fuel cost of the power only unit i at the beginning and end of the block k depends on its constant indices a_i, b_i, c_i and generated power, as modeled by Eqs. (14) and (15), respectively. As expected from Fig. 2, the total fuel cost of the thermal generation stations at T hours, FC_p , is calculated using (16). It is obvious that the operation cost of the thermal unit i is a quadratic function versus its power product at each time interval t . Hence, if the hourly fuel cost is calculated based on the power product of this unit, a complex nonlinear optimization problem should be solved to find the decision variables $u_{i,t}$ and $P_{i,t}$. Therefore, constraint (16) is used to calculate the operation cost of the thermal power plants in a linear manner. The parameters P_i^{min} and s_i^k are constant and calculated, respectively, while the term $\sum_{k=1}^n (s_i^k \times P_{i,t}^k)$ changes linearly with respect to $P_{i,t}^k$.

$$C_{i,ini}^k = a_i (P_{i,ini}^k)^2 + b_i P_{i,ini}^k + c_i; \forall k = 1: n \text{ and } \forall i \in \Omega_p \quad (14)$$

$$C_{i,fin}^k = a_i (P_{i,fin}^k)^2 + b_i P_{i,fin}^k + c_i; \forall k = 1: n \text{ and } \forall i \in \Omega_p \quad (15)$$

$$FC_p = \sum_{i \in \Omega_p, t \in T} \left\{ a_i (P_i^{min})^2 + b_i P_i^{min} + c_i + \sum_{k=1}^n (s_i^k \times P_{i,t}^k) \right\} \quad (16)$$

Each thermal unit i at hour t generates a limited power between its lower (P_i^{min}) and upper ($\bar{P}_{i,t}$) bounds, as (17). According to (18), when this power only unit shuts down at time $t + 1$ ($z_{i,t+1} = 1$), its maximum power generation capability at hour t will be restricted by its shut-down ramp rate as term $\bar{P}_{i,t} \leq SD_i \times z_{i,t+1}$. If this unit is on in time $t - 1$ ($u_{i,t-1} = 1$) and remains on at hour t , the power generation increase will be less than its ramp up rate RU_i , as expected from (19). Moreover, if it is off at time period $t - 1$ ($u_{i,t-1} = 0$) and starts up at hour t ($y_{i,t} = 1$), the value of the power generation increase will be smaller than its start-up index, SU_i . The binary decision variables $z_{i,t+1}$ and $y_{i,t}$ are used for modeling the start-up and shut-down of the power only unit i . According to (18), the upper time-dependent operating limit of the thermal unit i should be smaller than its power generation capacity, which means that $\bar{P}_{i,t} \leq P_i^{max}$. Additionally, if the thermal unit i is on at

hour t ($u_{i,t} = 1$) and shuts down at the next hour ($z_{i,t+1} = 1, P_{i,t+1} = 0$), its upper operating limit at hour t cannot be more than the shut-down index RU_i and hence $\bar{P}_{i,t} \leq SD_i$. Constraint (19) proves that when the thermal unit i has been started up at hour $t - 1$ ($u_{i,t-1} = 1$) and is going to remain on a time t , its power output ($P_{i,t}$) cannot be increased more than RU_i , that causes $P_{i,t} \leq P_{i,t-1} + RU_i \times u_{i,t-1}$. Similarly, if the unit i is off at the previous hour ($u_{i,t-1} = 0$) and is started up at time t ($y_{i,t} = 1$), $P_{i,t}$ cannot be more than its start-up value and $P_{i,t} \leq SU_i$.

$$P_{i,t} \leq \bar{P}_{i,t} \leq \bar{P}_{i,t}; \forall i \in \Omega_p \text{ and } \forall t \in T \quad (17)$$

$$\bar{P}_{i,t} \leq P_i^{max} (u_{i,t} - z_{i,t+1}) + SD_i \times z_{i,t+1} \quad (18)$$

$$P_{i,t} \leq P_{i,t-1} + RU_i \times u_{i,t-1} + SU_i \times y_{i,t} \quad (19)$$

As given by constraint (20), the electrical power output of the thermal unit i is equal to or greater than its minimum production capacity, $P_i^{min} \times u_{i,t}$. This means that if the unit i is on in time t , its power generation will be greater than P_i^{min} . In addition, while this generating unit is on at hours $t - 1$ and t , its output power will be larger than $P_{i,t-1} - RD_i \times u_{i,t-1}$. Note that RD_i represents its ramp down rate in MW per hour. Similarly, if this generation station is on in time $t - 1$ and shuts down at time interval t ($z_{i,t} = 1$), $P_{i,t-1}$ will be smaller than its shut-down limit, RU_i .

$$P_{i,t} \geq P_i^{min} \times u_{i,t}; \forall i \in \Omega_p \text{ and } \forall t \in T \quad (20)$$

$$P_{i,t-1} \geq P_{i,t-1} - RD_i \times u_{i,t-1} - SD_i \times z_{i,t}; \forall i \in \Omega_p \text{ and } \forall t \in T \quad (21)$$

In fact, the auxiliary binary decision variables $y_{i,t}$ and $z_{i,t}$ are used for modeling the start-up and shut-down status of the power only unit i in time t , as stated in constraint (22). Based on inequality constraint (23), the start-up and shut-down operations of the generating units don't occur simultaneously.

$$y_{i,t} - z_{i,t} = u_{i,t} - u_{i,t-1}; \forall i \in \Omega_p \text{ and } \forall t \in T \quad (22)$$

$$y_{i,t} + z_{i,t} \leq 1; \forall i \in \Omega_p \text{ and } \forall t \in T \quad (23)$$

Constraints (24)–(26) are applied on the optimization problem to consider the minimum up time (UT_i) of the thermal generation station i . It is assumed that U_i^0 refers to the duration of the period, which the power plant i has been on at the end of $t = 0$. If the power only unit i has been on in time $t = 0$, Eq. (24) can be considered for fewer hours than its minimum up time UT_i . According to (25), minimum up time limit is implemented to all subintervals with length UT_i . As shown in (26), for the last UT_i hours, the unit i should be committed until the last hour, while it is started at each hour of the mentioned set.

$$\sum_{t=1}^{\zeta_i} 1 - u_{i,t} = 0; \text{ where } \zeta_i = \min\{T, (UT_i - U_i^0)u_{i,t=0}\} \quad (24)$$

$$\sum_{t=k}^{k+UT_i-1} u_{i,t} \geq UT_i \times y_{i,k}; \forall k = \zeta_i + 1, \dots, T - UT_i + 1 \quad (25)$$

$$\sum_{t=k}^T (u_{i,t} - y_{i,t}) \geq 0; \forall k = T - UT_i + 2, \dots, T \quad (26)$$

If the i th producer has been off in time $t = 0$, constraint (27) will be imposed on the optimization problem for fewer hours than its minimum down time, DT_i . It is supposed that S_i^0 shows the duration of period, which the power plant i has been off at the end of $t = 0$. Inequality constraint (28) is used for all subintervals with DT_i hours. As obvious from (29), for the last $DT_i - 1$ hours, the thermal power generator i will be decommitted until the last hour, while it is turned off within the mentioned period.

$$\sum_{i=1}^{\psi_i} u_{i,t} = 0; \text{ where } \psi_i = \min\{T, (DT_i - S_i^0)(1 - u_{i,t=0})\} \quad (27)$$

$$\sum_{i=k}^{k+DT_i-1} (1 - u_{i,t}) \geq DT_i \times z_{i,k}; \forall k = \psi_i + 1, \dots, T - DT_i + 1 \quad (28)$$

$$\sum_{i=k}^T (1 - u_{i,t} - y_{i,t}) \geq 0; \forall k = T - DT_i + 2, \dots, T \quad (29)$$

The start-up cost (STC_p) and shut-down cost (SDC_p) of the thermal generating units over the T -hour study horizon are computed as (30) and (31), respectively. In which, st_i and sd_i are the start-up cost and shut-down cost of the power only unit i at hour t in terms of \$/hour. It should be mentioned that $y_{i,t}$ and $z_{i,t}$ are the binary decision variables representing the start-up and shut-down of the power generation unit i in time t .

$$STC_p = \sum_{i \in \Omega_p, t \in T} (st_i \times y_{i,t}) \quad (30)$$

$$SDC_p = \sum_{i \in \Omega_p, t \in T} (sd_i \times z_{i,t}) \quad (31)$$

3. Robust optimization of water-power nexus model

The robust optimization approach, which was proposed by Soyster [59], models the uncertainty without knowing any information on the uncertain variable. In this method, it is assumed that the function $w = f(\alpha, \beta)$ changes non-linearly for known values β and linearly for uncertain variable α . Moreover, its value is calculated as inequality constraint (32) supposing that the probability density and membership functions of the uncertain variable α are not available. where, $\bar{\alpha}$ is its forecasted value. In addition, the maximum change of the uncertain variable α from the forecasted one is shown with $\hat{\alpha}$. The maximization of the objective function w is formulated as (33), while it is subjected to (32). It is assumed that the objective profit function w should be maximized considering the inequality constraint (32) for uncertain parameter α , which varies from $\hat{\alpha} - \bar{\alpha}$ to $\hat{\alpha} + \bar{\alpha}$. In addition, the objective function w changes linearly with the uncertain parameter α , while it varies non-linearly for another known or pre-defined variable β .

$$|\alpha - \bar{\alpha}| \leq \hat{\alpha} \quad (32)$$

$$\max_{\beta} w = f(\alpha, \beta) \quad (33)$$

Because of the linear variations of the objective function w with respect to uncertain variable α , the non-linear optimization problem can be given by (32)–(35). The matrix $A(\beta)$ represents the coefficients of the uncertain variable α with component $a_i(\beta)$. Moreover, the matrix $B(\beta)$ consists of the known variables β . If the linear function $f(\alpha, \beta)$ is maximized as high as possible, the objective function w will be maximized, significantly.

$$w \leq f(\alpha, \beta) \quad (34)$$

$$f(\alpha, \beta) = A(\beta) \times \alpha + B(\beta) \quad (35)$$

Constraint (36) is formulated to maximize the auxiliary function $f(\alpha, \beta)$ and then the main objective w . The risk-constrained strategy should be able to find a good solution vector with the maximum value of objective function w and guarantee its global optimality for a given prediction error. If there is a forecast error about parameter α , this scenario will be optimum with great probability by adding γ_i as a robust counter term to (35).

$$f(\alpha, \beta) = A(\beta) \times \bar{\alpha} + B(\beta) - \max_{\gamma_i} \sum_i a_i(\beta) \times \hat{\alpha} \times \gamma_i \quad (36)$$

$$\sum_i \gamma_i \leq \Gamma \quad (37)$$

$$0 \leq \gamma_i \leq 1 \quad (38)$$

Therefore, the objective function (33) and the constraints (36)–(38) can be considered as (39) and (40). where, the t^{th} forecasted decision variable is introduced as $\bar{\alpha}_t$. The 3rd term of the relation (36), $\max_{\gamma_i} \sum_i a_i(\beta) \times \hat{\alpha} \times \gamma_i$, is expanded as the matrix form (39). The inequality constraints (37) and (38) are satisfied by (40).

$$\max_{\gamma_i} [a_1(\beta) \times \bar{\alpha}_1 \quad a_2(\beta) \times \bar{\alpha}_2 \quad \dots \quad a_n(\beta) \times \bar{\alpha}_n] \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_n \end{pmatrix} \quad (39)$$

$$\begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_n \end{pmatrix} = \begin{pmatrix} \Gamma \\ 1 \\ \vdots \\ 1 \end{pmatrix} \quad (40)$$

Hence, the dual form of the optimization problem (36)–(38) is formulated as follows:

$$\min_{\xi_i, \delta} \Gamma \delta + \sum_i \xi_i \quad (41)$$

$$\delta + \xi_i \geq a_i(\beta) \times \bar{\alpha}_i \quad (42)$$

By considering (41) in the base optimization problem, the maximum value of the objective function w is found in a way that minimum value of $f(\alpha, \beta)$ is equal to w . As expected from (41), (44) and (45), $f(\alpha, \beta)$ is maximized for minimizing $\Gamma \delta + \sum_i \xi_i$ as low as possible. The minimization problem (41) is satisfied by adding the negative term $-\Gamma \delta - \sum_i \xi_i$ to constraint (35) and maximizing it by (43) and (44) as high as possible.

$$\max_{\beta, \xi_i, \delta} w \quad (43)$$

$$w \leq f(\alpha, \beta) \quad (44)$$

$$f(\alpha, \beta) = A(\beta) \times \bar{\alpha} + B(\beta) - \Gamma \delta - \sum_i \xi_i \quad (45)$$

$$\delta + \xi_i \geq A(\beta_i) \times \bar{\alpha}_i \quad (46)$$

As given by (1), the aggregator of power only units and CWP generation plants desires that the daily profit, which is equal to the revenue obtained from selling electricity minus the operation cost, is maximum. If the uncertain value of the electricity market price at hour t is introduced as λ_t and it is assumed that the hourly energy prices at $\Gamma\%$ of the study horizon are unknown, the robust scheduling of the water-power hub system can be expressed as (48) and (49). The operational constraints (3)–(31), (48) and (49) have been considered as the optimization constraints. The proposed model can be implemented to each large-scale water-power nexus (not distribution-level grids), which the limits of the thermal generation stations and combined potable water and power generation units (constraints (3)–(31)) as well as the robust constraints (criteria (48) and (49)) are considered in the robust optimization problem.

$$\max_{\xi_i, \delta, P_{i,t}^t, P_{j,t}, W_{j,t}} w \quad (47)$$

Subject to:

$$w \leq \sum_{i \in T} \bar{\lambda}_i P_{i,t}^t - C_{cwp} - FC_p - STC_p - SDC_p - \Gamma \delta - \sum_i \xi_i \quad (48)$$

$$\delta + \xi_i \geq P_{i,t}^t \times \hat{\lambda}_i \quad (49)$$

and constraints (3)–(31).

4. Case study and discussions

In this section, a standard combined drinking water and power generation system is studied to prove the capability of the proposed method in modeling the uncertainty of the electricity market prices. This benchmark cogeneration network is linked with the local power system. It consists of five power only units and two CWP plants. The start-up, shut-down, ramp-up, and ramp-down characteristics of the thermal units as well as their power generation bounds are presented in Table 1 [58]. The constant factors, power and water generation capacities, and the power-water ratio of the CWP units are reported in Table 2 [58]. The presented robust optimization model can be implemented on each plant-level CWP and thermal units, while minimum and maximum levels of water and power generations, ramp up and down rates of thermal units, power-water ratio of CWP producers, and other constant factors reported in Tables 1 and 2 are available for these units.

A mixed integer non-linear programming problem is coded under general algebraic mathematical modeling system (GAMS) and solved by convex over and under envelopes for nonlinear estimation (COUENNE) [60]. The forecasted electricity market prices are presented in Fig. 3 [61]. These values are determined using the historical data and the price forecasting methods [62]. The lower and upper boundaries of the energy prices are estimated based on the predicted values and shown in Fig. 3. The maximum deviation of the actual energy price from the forecasted one at each hour t is reported in Table 3. Firstly, the budget of uncertainty, Γ , is considered to be equal to zero for solving the base optimization problem without modeling the market price uncertainty. The diverse values of the control parameter Γ are then intended for developing the risk-constrained self-scheduling problem. The numerical results of the mentioned cases are compared to show how the system operator makes the best decision on the performance of thermal units and CWP plants under the uncertainty of the energy market prices.

In the robust optimization of the water-energy nexus model, it is assumed that the budget of uncertainty changes from 0 to 1. This means that the electricity market prices in 0–100% of the whole study horizon may be different from the forecasted values. For example, if the budget of uncertainty, Γ , is equal to 0.5, the aggregator of the generating units desires the maximum daily profit, while the energy prices at hours $t = 1$ to $t = 12$ may change from the predicted ones. The profit objective function is obtained from solving the robust optimization problem (47) subjecting to (3)–(31), (48) and (49) under different risk-neutral and risk-seeker decision-making scenarios. Fig. 4 demonstrates how the degree of conservatism, Γ , affects the daily profit. For the first iteration, the budget of uncertainty is equal to 0. Hence, the uncertainty of the electricity prices is not modeled in this deterministic or risk-neutral strategy. Hence, \$1,018,335 benefit is attained by solving the base optimization approach under the forecasted energy prices. By considering 10 risk-aversion or robust scenarios with 0.1 step size from $\Gamma = 0.1$ to $\Gamma = 1$, it is proved that when the budget of uncertainty increases, lower profit will be achieved from selling electricity and vice versa. Because, the aggregator of CWP plants and thermal units wants to hedge himself from the risk of the lower electricity prices. Therefore, the distance of the optimal solution from the best operating point (higher profit) will increase. If the decision maker desires a higher profit, the degree of conservatism or robustness will decrease. For instance, if the budget of uncertainty is equal to 0.7, the actual energy prices may change at 70% of 24-h time interval. At other remaining hours, the forecast error is equal to zero and the uncertain price value, λ_t , is equal to the predicted price, $\hat{\lambda}_t$. Additionally, the gained profit will reduce from \$1,018,335 in the deterministic model to \$505,408 in the robust optimization problem with $\Gamma = 0.7$. In other words, there is a tradeoff between the degree of conservatism and the optimality of solutions.

The optimum generation schedules of the power only units 3 and 5

at three selected levels of control parameter Γ are obtained as Fig. 5. It is obvious from this figure that $\Gamma = 0$ models the risk-neutral or deterministic approach, while $\Gamma = 0.5$ and 1 are considered in two robust self-scheduling strategies. The sections (a) and (b) of Fig. 5 show the output electrical power of the thermal generating units 3 and 5 over the 24-h study period, respectively. As expected from objective function, when the combined water and power generation company desires the higher profit, the thermal power plants should generate more electricity and vice versa. In addition, when the Genco wants to be immune from the risk of the under-estimated energy market prices, he should consider the higher value for the budget of uncertainty. Therefore, it will be economic to generate lower electrical power by thermal power plants. As illustrated in Fig. 5, the blue and red curves of the sections (a) and (b) show the daily generation patterns of the power only units 1 and 2 in the cases of $\Gamma = 0.5$ and 1, respectively. It is demonstrated that by considering the uncertainty of the electricity prices in 50 and 100% of the whole study horizon, the value of the electrical power procured by the thermal units is equal to or less than that of obtained from solving the deterministic self-scheduling problem (1)–(31). Similar analyses can be provided for other power generating units. Therefore, the total profit attained by selling energy to the local electricity grid decreases from \$1,018,335 in the case of $\Gamma = 0$ to \$626,731 and \$401,809 in the cases of $\Gamma = 0.5$ and $\Gamma = 1$, respectively.

In Figs. 6 and 7, the best operating points of the CWP plants 1 and 2 under different budgets of uncertainty are displayed. This is an interesting result indicating that if it is desirable to make a robust or risk-aversion decision against the fluctuations of the electricity market prices, the CWP generating units should produce less electricity than their deterministic production schedules. Moreover, the output electrical power of the co-producer 1 at some hours such as $t = 1$ to $t = 6$ and $t = 19$ overlap in deterministic ($\Gamma = 0$) and robust ($\Gamma = 0.5$) modes. But, they are different at other hours. It is found that if the system operator desires to optimally schedule the water and power generating units under the variable energy prices ($\Gamma = 1$), the CWP units 1 and 2 should generate electricity according to the red dashed lines of the sections (a) and (b). These curves help the Genco to decide how the thermal and the CWP should be co-dispatched to minimize the risk of the lower energy prices and maximize the total profit. Additionally, if the electricity market prices may be different from the forecasted ones at the first 12 h ($\Gamma = 0.5$) and the entire time interval ($\Gamma = 1$), the aggregator can determine the amount of the electricity, which should be generated by the power only units and the co-production facilities at each hour.

As shown in Fig. 6, although the actual electricity rates differ from the predicted ones at $t = 1$ to $t = 12$ in the case of $\Gamma = 0.5$, the power production patterns of the combined cycles 1 and 2 are similar to their generation schedules obtained from solving the risk-neutral

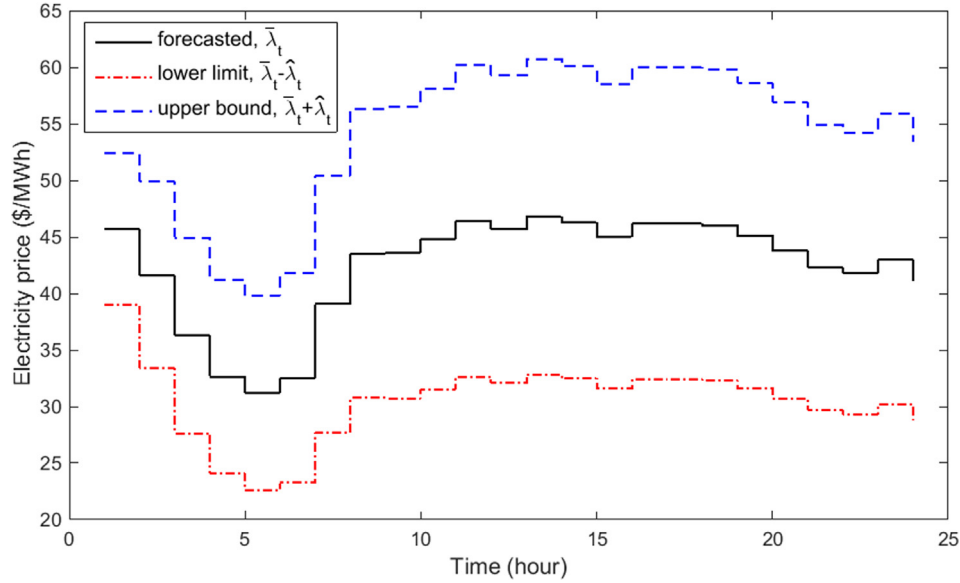
Table 1
The operational indices of the thermal generating units [58].

Constants	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$
a_i	0.039	0.028	0.051	0.032	0.046
b_i	15	12.6	14.3	13.9	13.5
c_i	25	49	15	105	72
sd_i	50.6	50.6	57.1	47.1	56.6
st_i	50.6	50.6	57.1	47.9	56.9
RU_i	16	64	12	104	56
RD_i	16	64	12	104	56
UT_i	0	4	0	5	4
DT_i	0	2	0	3	2
RU_i	35	130	30	240	110
SU_i	145	140	140	250	130
p_i^{min}	120	120	120	250	80
p_i^{max}	480	320	260	520	280
U_i^0	0	2	0	1	1
S_i^0	0	0	0	0	0

Table 2

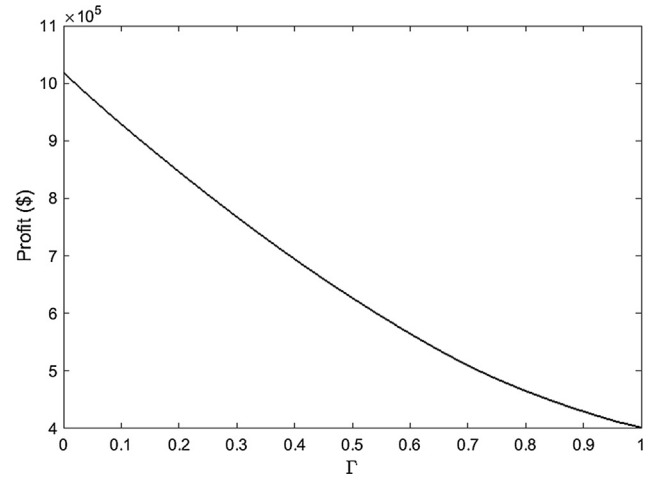
The power and water generation factors of the CWP units.

j	a_j	b_j	c_j	d_j	e_j	f_j	P_j^{min}	P_j^{max}	W_j^{min}	W_j^{max}	R_j^{min}	R_j^{max}
1	0.0434433	0.043546	0.037093	-1.106	-4.426	937.4	60	800	15	200	4	15
2	0.0327881	0.036305	0.04261	-1.475	-5.901	537.4	20	600	13	150	4	15

**Fig. 3.** The forecasted values, upper bound and lower limit of the electricity prices.**Table 3**The maximum value of the price deviation at each hour t , $\hat{\lambda}_t$.

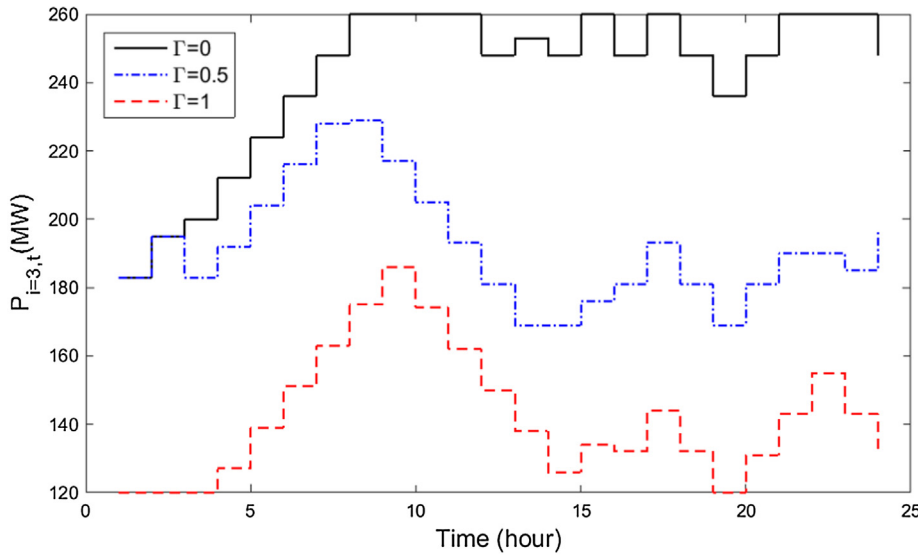
t	$\hat{\lambda}_t$
1	6.72
2	8.27
3	8.65
4	8.6
5	8.61
6	9.26
7	11.34
8	12.76
9	12.88
10	13.29
11	13.8
12	13.6
13	13.95
14	13.81
15	13.44
16	13.79
17	13.79
18	13.75
19	13.48
20	13.09
21	12.64
22	12.48
23	12.85
24	12.29

optimization problem ($\Gamma = 0$). In fact, there is no need to reduce the power generation of the combined water and electricity producers at these hours, while the electricity market prices change at $t = 1$ to $t = 12$. Fig. 7 proves that if the degree of conservatism increases, the total profit and the electrical power produced by the CWP units will decrease to hedge the Genco against the smaller electricity tariffs. But, the volume of the fresh water output of the cogeneration units may increase or decrease as appeared in parts (a) and (b) of Fig. 7, respectively.

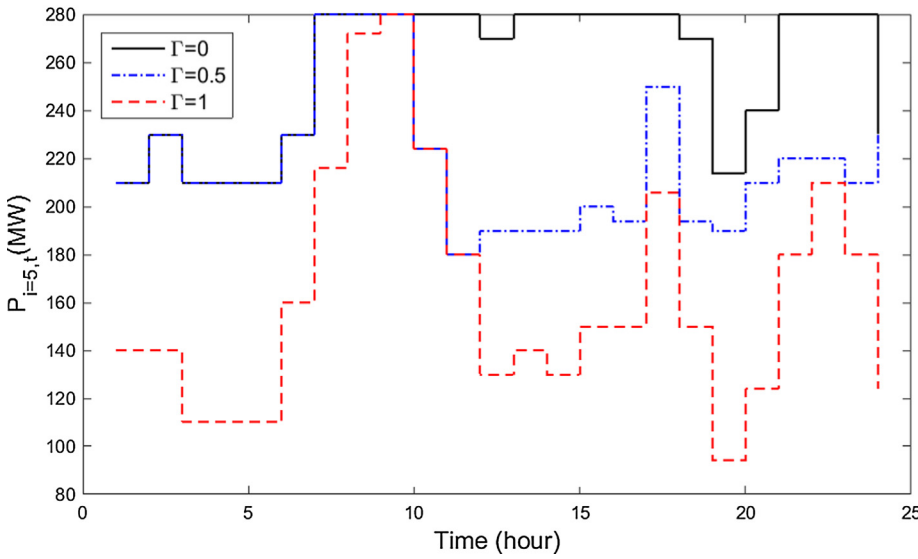
**Fig. 4.** The expected profit values versus the budget of uncertainty.

Depending on the power-water ratio of the CWP units, it will sometimes not be necessary to reduce the potable water production of all co-producers, when the budget of uncertainty increases. The total electricity and drinking water generated by the thermal units and CWP plants in three cases are represented in Fig. 8. It is revealed that while the aggregator considers the uncertainty of the electricity prices aiming to reduce the risk of the under-forecasted energy prices, the total electrical power and pure water produced by the generation stations will either decrease or not change. Therefore, the daily benefit achieved from trading energy with the main electricity network will reduce. These curves enable the system operator to know how the generation patterns of the producers will change at each hour, when the actual energy prices may be different from the forecasted values in 0–100% of the entire study period.

In this research, the profit objective function (1) was firstly



(a) Thermal unit 3

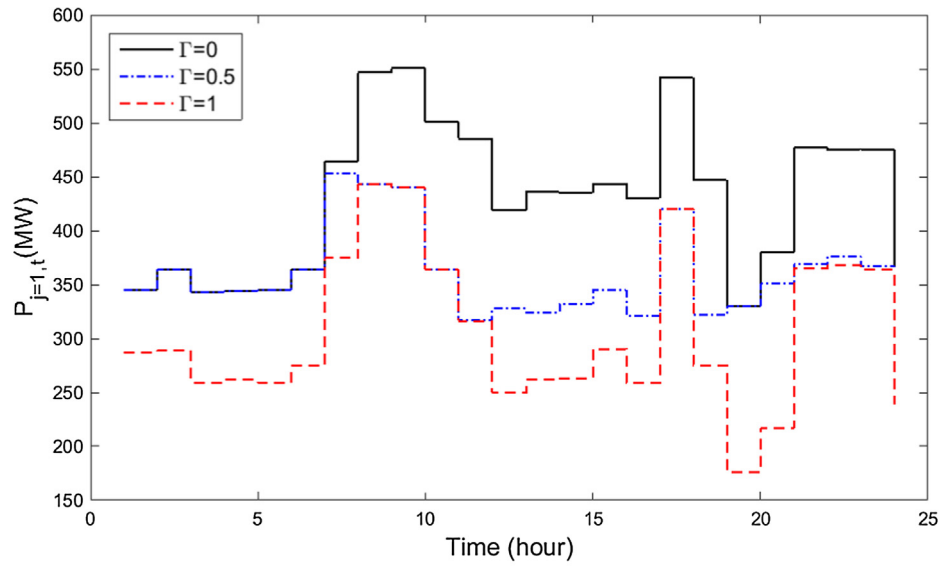


(b) Thermal unit 5

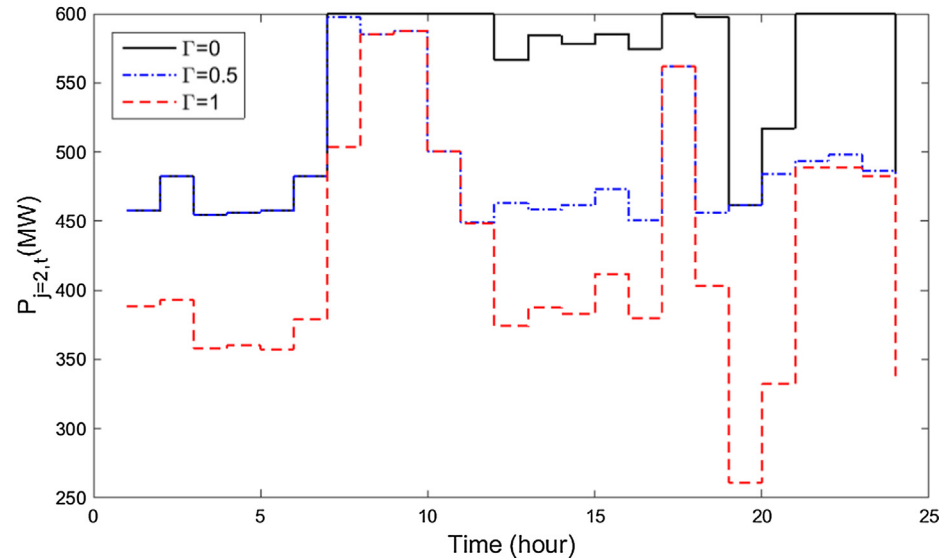
Fig. 5. The power generation of the thermal units 3 and 5 for various levels of Γ .

maximized taking into account the operational limits (2)–(31) to find the best generation schedules of the combined cycle units and thermal power plants during the sample day. In this case study, the uncertainty of the electricity market prices is not considered in the optimization process to determine the fresh water and power generation patterns of the producers before implementation of robust optimization algorithm. This scenario is called deterministic or risk-neutral because the fluctuations of the energy rates are not considered and the budget of uncertainty is equal to zero ($\Gamma = 0$). The forecasted values of the hourly electricity prices, shown in Fig. 3, are inserted in the deterministic profit maximization problem (1)–(31). In the second case study, the uncertainty associated with energy market prices is modeled using the robust optimization approach. Therefore, the degree of conservatism, Γ , is changed to be more than 0 and less than 1, while solving the risk-constrained short-term optimization framework (47)–(49) and (3)–(31). The maximum deviation of the actual energy price from the forecasted one at hour t , $\hat{\lambda}_t$, is considered as shown in Fig. 3. Day-ahead generation schedules of thermal and CWP units are then compared and discussed in deterministic and robust decision making procedures.

It is found that if the degree of conservatism increases, lower profit will be obtained from selling energy to the local power system. Because, the system operator desires to hedge himself from the risk of the lower energy rates. Hence, the distance of the optimal operating points of the water-power hub system from the best solution (the highest profit) will increase. If the aggregator of CWP and thermal units wants the greater profit, the budget of uncertainty or robustness factor will reduce, as proved in Fig. 4. If the cogeneration company seeks the greater profit, the power only units should generate more electricity and vice versa. If he wants to be immune against the risk of the under-estimated electricity tariffs, he should intend the higher degrees of conservatism. In two robust cases with $\Gamma = 0.5$ and $\Gamma = 1$, the value of the electrical power procured by the thermal units is equal to or less than that found in deterministic or risk-neutral mode. In addition, when the decision maker optimally schedules the pure water and power production units under the severe-uncertain condition ($\Gamma = 1$), the Genco will be able to decide how the co-generation units and power plants should be operated to minimize the risk of the lower power market prices and maximize the daily net profit. When the energy rates are different from the



(a) Co-producer 1



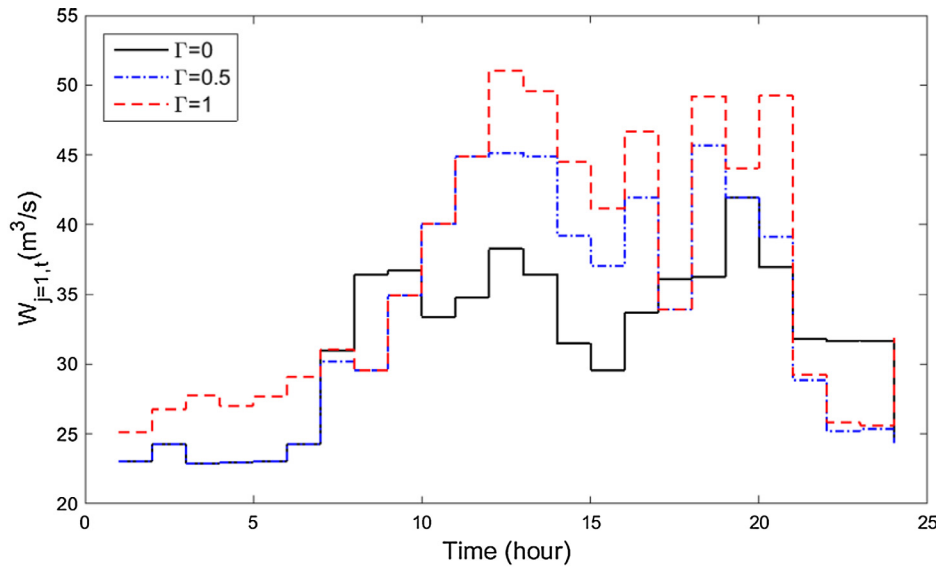
(b) Co-generator 2

Fig. 6. Daily electricity production of CWP plants 1 and 2 in risk-neutral and risk-aversion optimization approaches.

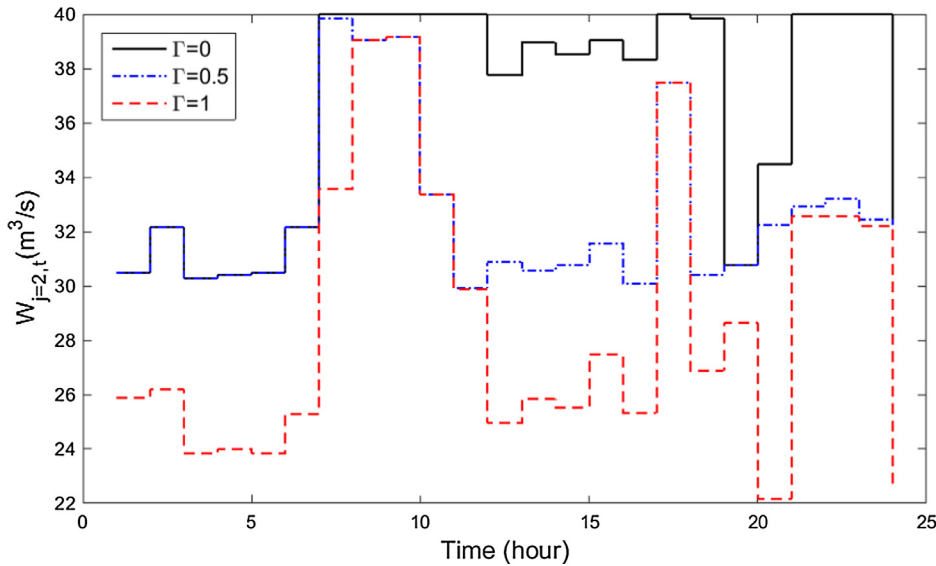
forecasted ones at the first 12 h ($\Gamma = 0.5$) and the whole study horizon ($\Gamma = 1$), the system operator will be able to determine the hourly electricity and desalinated water generation schemes of the power plants and the CWP generating units under the worst cases. An increase in the degree of conservatism causes the daily profit and the power output of the CWP producers decrease in order to hedge the cogeneration company from the under-estimated energy prices. While considering the uncertainty of the power market prices and minimizing the risk of their lower values, the power and potable water outputs of the generating units either decrease or not change, which causes the daily profit gained from trading electrical power with upstream energy network decreases. If the real-time electricity prices differ from the forecasted ones in 0 to 100% of the whole time interval, the cogeneration company can find how the power and drinkable water generation schemes of the generating units should be changed.

5. Conclusion

In this paper, the day-ahead robust self-scheduling of the conventional combined potable water and power generation system was presented. It was assumed that the power only units and the CWP plants were co-dispatched to supply the freshwater and energy demands of the benchmark network. The main objective of the CWP units is to generate electricity, but the waste heat of the high-temperature flue gases is recovered for seawater purification. In other words, the freshwater was considered as a byproduct. An aggregator of thermal units and co-producers was responsible for maximization of daily profit obtained from selling electricity. Meanwhile, the uncertain energy prices affect the expected revenue and the total profit gained over the study period. Therefore, a robust non-linear programming methodology was proposed to incorporate the variable nature of the electricity market prices in the water-energy nexus model. Firstly, a deterministic or risk-neutral optimization problem was solved to find the optimal generation



(a) CWP unit 1



(b) CWP unit 2

Fig. 7. Robust fresh water generation pattern of CWP units in various levels of Γ .

schedules and the greatest level of the daily profit. Afterward, the budget of uncertainty was changed from 0 to 100% in order to investigate the system performance under the fluctuations of the electricity rates. It is found that if the budget of uncertainty increases, the aggregator of CWP plants and thermal units will gain the lower profit to hedge himself from the risk of the lower market prices. If the decision maker desires a higher profit, the degree of conservatism or robustness will decrease. Moreover, the power and drinkable water production patterns are compared in risk-neutral and robustness iterations to know how these units should be operated when the actual electricity prices are different from the forecasted values at $0 \leq \Gamma \leq 1\%$ of the whole study horizon. The proposed approach enables the system operator to make the robust decisions and schedule the potable water and power generation of the conventional thermal and CWP units against the undesirable energy prices. As the future trends, the natural gas-water-power integrated networks should be studied using the optimal load flow analysis and considering the interconnection between the gas-fired Brayton power generation cycle, gas consumed by the gas load centers,

gas transmitted via the pipelines, gas extracted from suppliers/gas fields or wells, and the natural gas consumption in thermally-driven desalination plants. In the natural gas-fresh water-power nexus model, the economic benefit of the power to gas technology, that converts the electrical power to the gas fuel, should also be discussed. In addition, the risk-seeker decision making strategy can be implemented to the water-power nexus model as well as the gas, water, energy hybrid grids to benefit from the overestimated electricity market prices and the under-estimated values of the gas/water/power demands.

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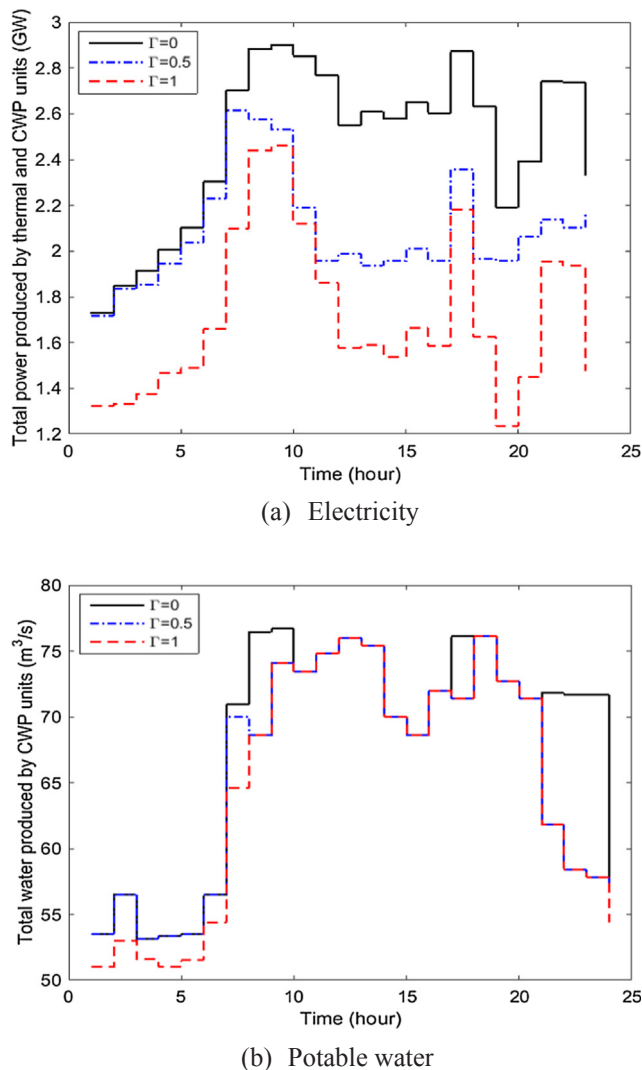


Fig. 8. Total electricity and fresh water generated by CWP units and thermal power plants for deterministic and robust solutions.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2019.114258>.

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